# Frequency Response of the RC-Coupled BJT Amplifier

Low-, Mid- and High-Frequency Analysis — Built From Reactance Alone

A self-contained study guide. No Fourier series, no Laplace transforms, no transfer functions in $s$ — only Ohm's law, capacitive reactance, and the small-signal model you already know.

---

**What you already know, and what this adds**

**Assumed:** DC biasing of the BJT (voltage-divider bias, finding $I_E$, $V_{CE}$), and mid-band small-signal analysis using $r_e$, $g_m$, $r_\pi$ and $\beta$ — i.e. you can already compute $A_v = -g_m(R_C \parallel R_L)$ with every capacitor treated as a perfect short.

**Added here:** what happens when the capacitors are *not* perfect shorts. That single relaxation is the whole subject. You will end up able to draw and justify the gain-versus-frequency curve of any RC-coupled stage.

**Deliberately avoided:** Laplace transforms, $s$-domain transfer functions, poles and zeros as formal objects, Fourier analysis. Everything below is ordinary steady-state sinusoidal circuit analysis at one frequency at a time.

## Contents

# 1 The Toolkit: Four Facts You Need First

Since we are not using transforms, we need a small set of tools. All four are things you can verify with a calculator.

## 1.1 Fact 1 — A capacitor's opposition to current depends on frequency

For a sinusoidal signal of frequency $f$, a capacitor $C$ opposes current with a **reactance**

$$X_C = \frac{1}{2\pi f C} \qquad \text{[ohms]} \tag{1}$$

This is the only new "component law" in the entire document. Notice the frequency in the denominator:

| Frequency | Reactance | Capacitor behaves as |
|---|---|---|
| $f \to 0$ (DC) | $X_C \to \infty$ | an **open circuit** |
| $f$ small | $X_C$ large | a large resistor |
| $f$ large | $X_C$ small | a small resistor |
| $f \to \infty$ | $X_C \to 0$ | a **short circuit** |

To feel the numbers, here is $X_C$ for two capacitors that both appear in our amplifier — one "large" external capacitor and one "small" internal parasitic:

| $f$ | 10 Hz | 1 kHz | 100 kHz | 1 MHz | 10 MHz |
|---|---|---|---|---|---|
| $X_C$ for $C = 10\,\mu$F | 1.6 kΩ | 16 Ω | 0.16 Ω | 0.016 Ω | 0.0016 Ω |
| $X_C$ for $C = 30$ pF | 531 MΩ | 5.3 MΩ | 53 kΩ | 5.3 kΩ | 531 Ω |

**Read that table again — it is the whole story**

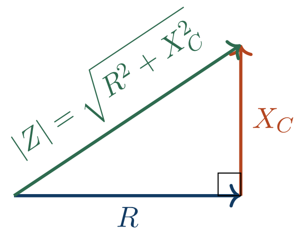
In the middle of the table (1 kHz to 100 kHz), the 10 $\mu$F capacitor is essentially a piece of wire (16 Ω and below) *and* the 30 pF capacitor is essentially not there (5.3 MΩ and above). That is the mid-band.
The two ends of the table are where one family or the other stops cooperating. The low-frequency analysis is about the big capacitors failing to be shorts. The high-frequency analysis is about the small capacitors failing to be opens.

### 1.2 Fact 2 — The "10:1 rule" for deciding short versus open

In practice we say a capacitor *is* a short if its reactance is at least ten times smaller than the resistance it sits next to, and *is* an open if its reactance is at least ten times larger. Between those two, it is genuinely a capacitor and you must keep it in the algebra. This rule is why the frequency axis splits so cleanly into three regions.

### 1.3 Fact 3 — The magnitude of a two-element divider

Every network in this document reduces to one resistor and one capacitor in series, with the output taken across one of them. You do not need phasor algebra to get the magnitude; you need one rule: *a resistance and a reactance in series add like the two legs of a right triangle.*




So the total opposition of $R$ and $C$ in series is $|Z| = \sqrt{R^2 + X_C^2}$, and the fraction of the input voltage appearing across the resistor is

$$\left|\frac{V_R}{V_{in}}\right| = \frac{R}{\sqrt{R^2 + X_C^2}} \qquad \text{and across the capacitor} \qquad \left|\frac{V_C}{V_{in}}\right| = \frac{X_C}{\sqrt{R^2 + X_C^2}}. \tag{2}$$

(A full derivation, if you want it, is in Appendix A. But you can also simply accept the right-triangle picture — it is exactly what "90° out of phase" means geometrically.)

### 1.4 Fact 4 — Decibels and logarithmic axes

Voltage gain in decibels is $A_{dB} = 20\log_{10}|A_v|$. Useful landmarks:

| $\lvert A_v\rvert$ | 0.707 | 1 | 2 | 10 | 100 | 1000 |
|---|---|---|---|---|---|---|
| $A_{dB}$ | $-3$ dB | 0 dB | 6 dB | 20 dB | 40 dB | 60 dB |

We use dB and a logarithmic frequency axis for two practical reasons. First, the interesting behaviour spans from a few hertz to several megahertz — six or seven **decades** (factors of ten) — and a linear axis cannot show that. Second, gains that *multiply* become dB values that *add*, so the effect of several capacitors can be built up by simple addition. That is the only reason this plot is drawn the way it is; there is nothing deeper going on.

**The $-3$ dB point**

$-3$ dB corresponds to $|A_v|$ falling to $1/\sqrt{2} = 0.707$ of its mid-band value. This is the universally agreed definition of the edge of the useful band. It is chosen because at 0.707 of the voltage, the *power* delivered to a load is exactly half — hence the alternative name **half-power point**. The frequencies where this happens are the **cutoff frequencies** $f_L$ (lower) and $f_H$ (upper), and $\text{BW} = f_H - f_L$ is the **bandwidth**.

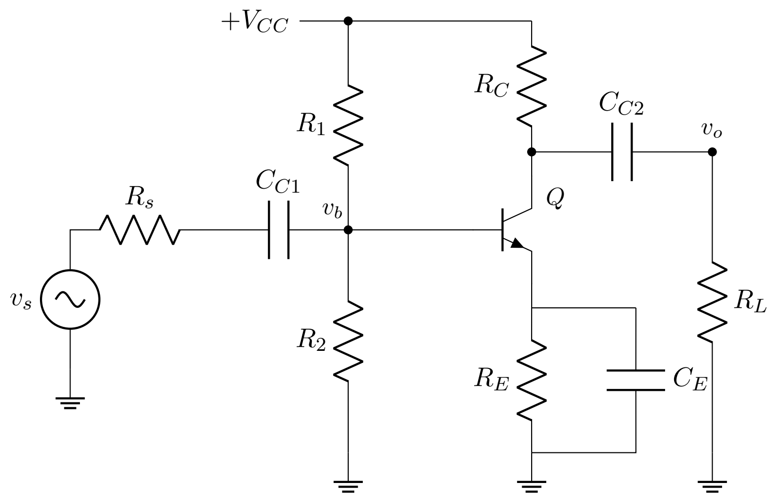



Figure 1: The common-emitter RC-coupled amplifier. $C_{C1}$ and $C_{C2}$ are the coupling capacitors; $C_E$ is the emitter bypass capacitor. Two more capacitors — invisible on this schematic — live inside the transistor.

## 2 The Circuit and Its Five Capacitors

### 2.1 Why each capacitor is there in the first place

$C_{C1}$**, input coupling.** Passes the signal from the source into the base, while blocking DC so that the source cannot disturb the base bias voltage $V_B$ set by $R_1$ and $R_2$. Without it, a source with a DC path to ground would drag $V_B$ down and destroy the operating point.

$C_{C2}$**, output coupling.** Passes the amplified signal to the load while blocking the DC collector voltage $V_C$, so that $R_L$ neither sees a DC offset nor loads the collector at DC.

$C_E$**, emitter bypass.** $R_E$ is needed for *DC* stability of the bias point, but at signal frequencies it introduces heavy negative feedback that would reduce the gain from $-R'_C/r_e$ down to $-R'_C/(r_e+R_E)$ — a catastrophic loss. $C_E$ short-circuits $R_E$ for AC only, giving us DC stability and full AC gain simultaneously.

$C_\pi$ **(or** $C_{be}$**), internal.** The base–emitter junction is forward biased, so it has both a depletion capacitance and a (usually dominant) diffusion capacitance from stored minority charge. Typically tens of picofarads. Nobody put it there; it is a consequence of physics.

$C_\mu$ **(or** $C_{bc}$**), internal.** The base–collector junction is reverse biased, so it has a depletion capacitance. Typically a few picofarads. Small, but as we will see, disproportionately destructive.

### 2.2 The organising insight: series versus shunt

Sort the five capacitors not by their value but by *how they sit in the signal path.*

| Capacitor | Size | Position | Failure mode |
|---|---|---|---|
| $C_{C1}$, $C_{C2}$, $C_E$ | $\mu$F | *in series* with signal (or across $R_E$) | fails as an open at **low** $f$ |
| $C_\pi$, $C_\mu$ | pF | *shunting* the signal to ground | fails as a short at **high** $f$ |

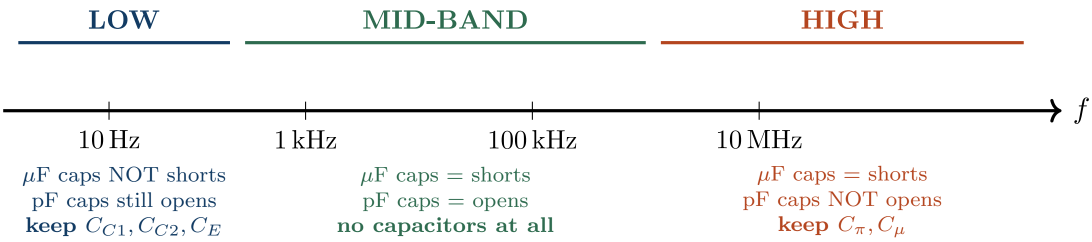
**Series capacitors kill low frequencies; shunt capacitors kill high frequencies**

A capacitor *in series* with the signal must become a short for the signal to get through. It does so at high frequency and fails to do so at low frequency. Hence it removes low frequencies.
A capacitor *in shunt* (from signal node to ground) must become an open for the signal to survive. It does so at low frequency and fails to do so at high frequency. Hence it removes high

frequencies.
The bypass capacitor $C_E$ is a special case: it is in shunt with $R_E$, but $R_E$ is a *gain-reducing* element, so bypassing it *increases* gain. It therefore behaves like the series capacitors — it hurts the low end.

## 3 Why We Split the Axis into Three Bands

Because the two families of capacitors are separated by six orders of magnitude in value, they act in completely separate regions of the frequency axis. This lets us do three *simple* analyses instead of one horrible one.




**The three equivalent circuits**

**Mid-band:** all large capacitors → short, all small capacitors → open. Nothing is left but resistors. This is the familiar analysis you already know, and it gives the reference gain $A_m$.
**Low-frequency band:** small capacitors are still open (their reactance is enormous), so ignore them. Keep the three large capacitors. Gain is *below* $A_m$.
**High-frequency band:** large capacitors are firmly shorts, so replace them by wires. Keep the two small capacitors. Gain is *below* $A_m$.

Notice that we never analyse all five capacitors at once. At every frequency, at most one family matters. This is not laziness — it is a genuine and accurate simplification, justified by the six-decade gap between $\mu$F and pF.

## 4 Mid-Band Analysis: The Reference Gain

With every capacitor replaced by a short or an open, the small-signal circuit is purely resistive.

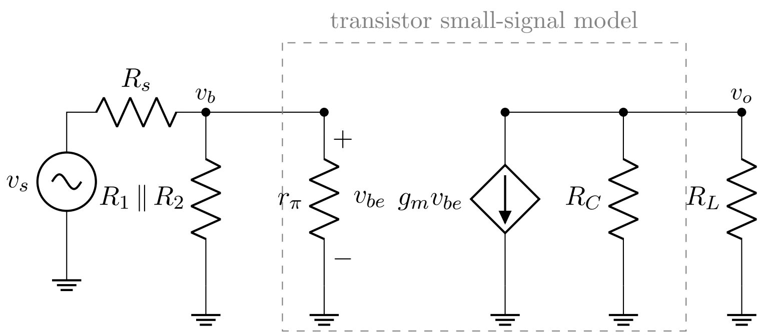



Figure 2: Mid-band small-signal equivalent. Every capacitor has vanished, leaving a purely resistive network. Note that the input side and output side are joined only through the dependent source $g_m v_{be}$ — there is no direct wire between base and collector once $C_\mu$ is removed.

The results, which you already know:

$$R'_C = R_C \parallel R_L \tag{3}$$

$$A_m = \frac{v_o}{v_b} = -g_m R'_C = -\frac{R'_C}{r_e} \tag{4}$$

$$R_i = R_1 \parallel R_2 \parallel r_\pi \tag{5}$$

$$A_{vs} = \frac{v_o}{v_s} = A_m \cdot \frac{R_i}{R_i + R_s} \tag{6}$$

The minus sign is the 180° phase inversion of the common-emitter stage. Throughout the rest of this document, $A_m$ is the fixed yardstick: every capacitor can only *reduce* the gain below $|A_m|$, never raise it. **The mid-band gain is flat** because it contains no frequency-dependent element whatsoever — that is precisely why the middle of the curve is a horizontal line.

### 4.1 Running numerical example

We will carry one set of numbers through the entire document.

| | | | |
|---|---|---|---|
| $V_{CC} = 12\,\text{V}$ | $R_1 = 40\,\text{k}\Omega$ | $R_2 = 10\,\text{k}\Omega$ | $R_C = 4\,\text{k}\Omega$ |
| $R_E = 1.5\,\text{k}\Omega$ | $R_L = 4\,\text{k}\Omega$ | $R_s = 600\,\Omega$ | $\beta = 100$ |
| $C_{C1} = 1\,\mu\text{F}$ | $C_{C2} = 1\,\mu\text{F}$ | $C_E = 20\,\mu\text{F}$ | |
| $C_\pi = 30\,\text{pF}$ | $C_\mu = 4\,\text{pF}$ | $C_{wiring} = 10\,\text{pF}$ | $V_T = 26\,\text{mV}$ |

**Bias:** $V_B = 12 \cdot \frac{10}{50} = 2.4\,\text{V}$, $V_E = 1.7\,\text{V}$, $I_E = 1.7/1.5\text{k} = 1.13\,\text{mA}$.
**Small signal:** $r_e = 26/1.13 = 23\,\Omega$, $g_m = 43.6\,\text{mS}$, $r_\pi = \beta r_e = 2.29\,\text{k}\Omega$.
**Mid-band:** $R'_C = 4 \parallel 4 = 2\,\text{k}\Omega$, so

$$A_m = -\frac{2000}{23} = \mathbf{-87.2} \;\; (38.8\,\text{dB}), \qquad R_i = 40\text{k} \parallel 10\text{k} \parallel 2.29\text{k} = 1.78\,\text{k}\Omega,$$

$$A_{vs} = -87.2 \times \frac{1.78}{1.78 + 0.6} = \mathbf{-65.2} \;\; (36.3\,\text{dB}).$$

## 5 Low-Frequency Analysis

Here the small internal capacitors have reactances in the megohms — utterly negligible — so we delete them. We keep $C_{C1}$, $C_{C2}$ and $C_E$, and we handle them **one at a time**, assuming the other two are still shorts. Each one produces its own cutoff frequency.

### 5.1 The universal building block: the high-pass RC section

Every low-frequency effect in this amplifier is the same circuit in disguise: a capacitor in series, a resistance to ground, output taken across the resistance.

Using Eq. (2), the fraction of $v_i$ appearing across $R$ is

$$\left|\frac{v_o}{v_i}\right| = \frac{R}{\sqrt{R^2 + X_C^2}} = \frac{1}{\sqrt{1 + (X_C/R)^2}}.$$

Now substitute $X_C = 1/(2\pi f C)$ and *define* the frequency at which $X_C = R$:

$$\boxed{f_c = \frac{1}{2\pi RC}} \tag{7}$$

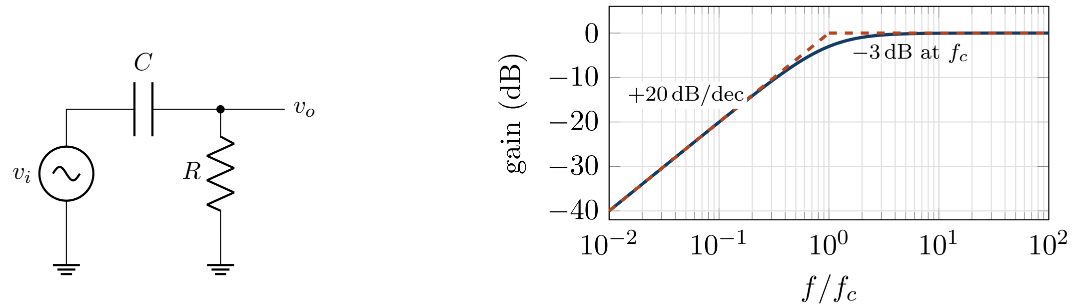



Figure 3: The high-pass section and its response. Solid: true magnitude. Dashed: the straight-line (asymptotic) approximation.

Then $X_C/R = f_c/f$, and the result collapses to the single formula you will use five times in this document:

$$\left|\frac{v_o}{v_i}\right| = \frac{1}{\sqrt{1+\left(\frac{f_c}{f}\right)^2}} \tag{8}$$

Read off its three behaviours:

- $f \gg f_c$: the fraction $\to 1$. Full signal passes. *This is the mid-band.*
- $f = f_c$: the fraction $= 1/\sqrt{2} = 0.707$, i.e. $-3$ dB. *This is the cutoff.*
- $f \ll f_c$: the fraction $\to f/f_c$. Halving the frequency halves the output. On a log–log plot this is a straight line of slope +20 dB per decade. *This is the roll-off.*

**Where the 20 dB/decade slope comes from — no calculus needed**

Well below cutoff, $X_C \gg R$, so almost all of $v_i$ is dropped across the capacitor and only the small fraction $R/X_C$ reaches the output. Since $X_C \propto 1/f$, that fraction is proportional to $f$. So if you divide the frequency by 10, the output divides by 10, which is $-20$ dB. One capacitor $\Rightarrow$ one factor of $f \Rightarrow$ 20 dB per decade. Two capacitors acting together give 40 dB/decade, three give 60 dB/decade. That is all the ”slope” ever means.

**The recipe for finding $f_c$**

For each capacitor: **(1)** short the independent sources, **(2)** short the *other* capacitors, **(3)** look into the two terminals of this capacitor and compute the total resistance $R_{eq}$ it sees, **(4)** $f_c = 1/(2\pi R_{eq} C)$. That is the entire procedure.

### 5.2 Cutoff due to the input coupling capacitor $C_{C1}$

Looking out from $C_{C1}$'s terminals: on one side the source resistance $R_s$, on the other the amplifier input resistance $R_i = R_1 \parallel R_2 \parallel r_\pi$. They are in series with the capacitor.

$$\boxed{f_{L1} = \frac{1}{2\pi(R_s + R_i)\,C_{C1}}} \tag{9}$$

$$\text{Example: } f_{L1} = \frac{1}{2\pi(600+1783)(1\times10^{-6})} = \mathbf{66.8}\,\text{Hz}$$

### 5.3 Cutoff due to the output coupling capacitor $C_{C2}$

Looking out from $C_{C2}$'s terminals: on one side, the resistance seen at the collector, which is $R_C$ (the dependent current source is an ideal open circuit, so $r_o$ aside, nothing else appears); on the other side, $R_L$.

$$\boxed{f_{L2} = \frac{1}{2\pi(R_C + R_L)\,C_{C2}}} \tag{10}$$

$$\text{Example: } f_{L2} = \frac{1}{2\pi(4000 + 4000)(1 \times 10^{-6})} = \mathbf{19.9}\,\text{Hz}$$

### 5.4 Cutoff due to the bypass capacitor $C_E$ — the awkward one

This is where students lose marks, so go slowly. $C_E$ sits from the emitter to ground. The resistance it sees is $R_E$ in parallel with *the resistance looking into the emitter terminal of the transistor.*

Looking into the emitter, you see $r_e$ plus whatever is in the base circuit — but divided by $(\beta+1)$, because the emitter current is $(\beta + 1)$ times the base current, so any base-side resistance appears $(\beta + 1)$ times smaller when viewed from the emitter.

$$R_{eq} = R_E \parallel \left(r_e + \frac{R_s'}{\beta + 1}\right), \qquad R_s' = R_s \parallel R_1 \parallel R_2 \tag{11}$$

$$\boxed{f_{LE} = \frac{1}{2\pi R_{eq} C_E}} \tag{12}$$

$$R_s' = 600 \parallel 40\text{k} \parallel 10\text{k} = 558\,\Omega, \qquad r_e + \frac{558}{101} = 23 + 5.5 = 28.5\,\Omega,$$

$$R_{eq} = 1500 \parallel 28.5 = 27.9\,\Omega, \qquad f_{LE} = \frac{1}{2\pi(27.9)(20 \times 10^{-6})} = \mathbf{284.8}\,\text{Hz}$$

**Why $C_E$ almost always dominates — and why it must be huge**

Look at the resistance: $27.9\,\Omega$. It is tiny, because $R_E$ is in parallel with the very low resistance looking into the emitter. A small $R_{eq}$ means a *high* $f_c$. So even a $20\,\mu$F capacitor gives a cutoff near 285 Hz, while $1\,\mu$F coupling capacitors give cutoffs of only 20–67 Hz. **The bypass capacitor is almost always the limiting element at the low end**, and this is why real amplifiers use enormous electrolytics (47–470 $\mu$F) there while 0.1–1 $\mu$F suffices for coupling. To move our example's bypass cutoff down to 60 Hz you would need $C_E = 1/(2\pi \times 27.9 \times 60) = 95\,\mu$F — so you would fit $100\,\mu$F.

**$C_E$ does not roll the gain off to zero**

The coupling capacitors are in series with the signal, so as $f \to 0$ they genuinely block everything and the gain goes to zero. $C_E$ is different. As $f \to 0$, $C_E$ simply stops working and $R_E$ becomes fully un-bypassed, leaving the finite gain

$$A_{v,\text{low}} = -\frac{R_C'}{r_e + R_E} = -\frac{2000}{23 + 1500} = -1.31 \;\; (2.4\,\text{dB}),$$

compared with $-87.2$ (38.8 dB) in mid-band. So $C_E$ produces a **step down of about** 36 dB, not an endless roll-off. It stops falling once $X_{C_E}$ has grown past $R_E$ itself, which happens below $f = 1/(2\pi R_E C_E) = 5.3$ Hz in our example. Below that, $C_E$'s contribution is flat again and only the coupling capacitors keep pulling the curve down.

### 5.5 Which cutoff wins?

Each capacitor pulls the gain down below its own $f_c$. Going down in frequency from mid-band, the *first* thing you hit is the **highest** of the three cutoffs. That one defines the lower $-3$ dB frequency of the amplifier.

$$f_L \approx \max\left(f_{L1},\, f_{L2},\, f_{LE}\right) \tag{13}$$

$$\text{Example: } f_L \approx \max(66.8,\ 19.9,\ 284.8) = \mathbf{285}\,\text{Hz}$$

(The exact value including the mild contribution of the other two is 302 Hz — the "max" rule is a good engineering estimate, typically a few percent optimistic.)

# 6 High-Frequency Analysis

Now flip to the other end. The $\mu$F capacitors have reactances of milliohms; they are perfect wires and we replace them with short circuits. What remains are the two internal transistor capacitances, plus wiring and load capacitance, all of which *shunt* the signal to ground.

## 6.1 The universal building block: the low-pass RC section

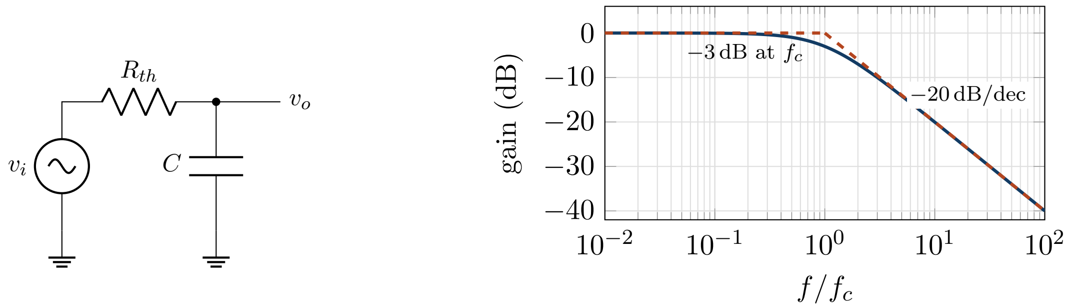



Figure 4: The low-pass section and its response — the mirror image of Figure 3.

Now the output is taken across the *capacitor*, so from Eq. (2),

$$\left|\frac{v_o}{v_i}\right| = \frac{X_C}{\sqrt{R_{th}^2 + X_C^2}} = \frac{1}{\sqrt{1 + (R_{th}/X_C)^2}}, \qquad f_c = \frac{1}{2\pi R_{th} C},$$

$$\left|\frac{v_o}{v_i}\right| = \frac{1}{\sqrt{1 + \left(\dfrac{f}{f_c}\right)^2}} \tag{14}$$

Identical in structure to Eq. (8), but with $f$ and $f_c$ swapped. Same $-3$ dB point, same 20 dB/decade slope, mirrored. Above cutoff, $X_C \ll R_{th}$, so the capacitor is stealing the signal: the fraction reaching the output is $X_C/R_{th} \propto 1/f$, hence $-20$ dB per decade.

Here $R_{th}$ is the Thevenin resistance seen by the capacitor *looking back into the circuit* from the node it shunts.

## 6.2 The high-frequency small-signal model

$C_\pi$ is simply in shunt at the base node, so it is already in low-pass form. $C_\mu$ is the problem: it connects the base to the collector, and the collector voltage is a large, *inverted* version of the base voltage. We need a way to convert that bridging capacitor into ordinary shunt capacitors. That method is the Miller effect.

## 6.3 The Miller effect — derived with only Ohm's law

Consider any impedance $Z$ bridging the input and output of an amplifier whose voltage gain is $A_v = v_o/v_i$ (for our stage, $A_v$ is a large negative number).

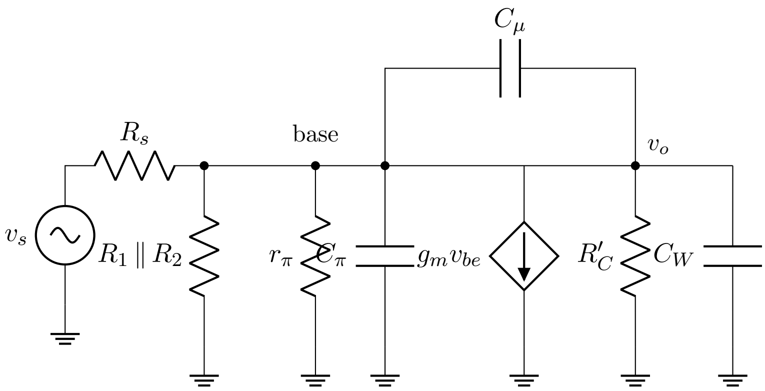
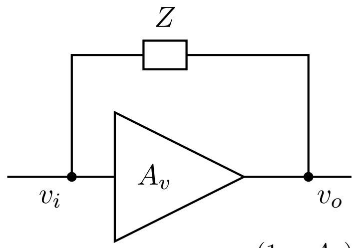



Figure 5: High-frequency hybrid-$\pi$ model. All coupling and bypass capacitors are now short circuits. $C_\mu$ bridges input and output — this is what makes the analysis interesting. $C_W$ lumps wiring and load capacitance.




$$\text{current } i = \frac{v_i - v_o}{Z} = \frac{v_i(1 - A_v)}{Z}$$

The current drawn from the input node through $Z$ is

$$i = \frac{v_i - v_o}{Z} = \frac{v_i - A_v v_i}{Z} = \frac{v_i(1 - A_v)}{Z}.$$

But the input node cannot tell the difference between this and an impedance connected straight to ground that draws the same current. That equivalent grounded impedance is

$$Z_M = \frac{v_i}{i} = \frac{Z}{1 - A_v}.$$

For a capacitor, $Z = 1/(2\pi f C)$ in magnitude, and dividing an impedance by $(1 - A_v)$ is the same as *multiplying the capacitance* by $(1 - A_v)$:

$$\boxed{C_M = C\,(1 - A_v) = C\,(1 + |A_v|)} \qquad \text{(for inverting } A_v\text{)} \tag{15}$$

**What the Miller effect physically means**

When the base rises by 1 mV, the collector *falls* by 87 mV. So the voltage across $C_\mu$ swings by 88 mV, not 1 mV. To produce an 88 mV swing across it, the source must supply 88 times as much charge as it would to a grounded capacitor. The input therefore sees a capacitor that appears 88 times bigger than it really is. A harmless 4 pF becomes a crippling 353 pF.
**The high gain of the stage is what destroys its own bandwidth.** This is the single most important idea at the high-frequency end.

A similar argument at the output node gives the output-side Miller capacitance $C_{M,out} = C_\mu(1 + 1/|A_v|) \approx C_\mu$, which is almost negligible. So $C_\mu$ is reflected almost entirely to the input.

### 6.4 Input-side cutoff

$$C_{in} = C_\pi + C_\mu(1 + |A_m|), \qquad R_{th,in} = R_s \parallel R_1 \parallel R_2 \parallel r_\pi, \qquad \boxed{f_{H1} = \frac{1}{2\pi R_{th,in} C_{in}}} \tag{16}$$

$$\begin{aligned} C_{in} &= 30 + 4(1 + 87.2) = 30 + 353 = 383\,\text{pF} \\ R_{th,in} &= 600 \parallel 40\text{k} \parallel 10\text{k} \parallel 2.29\text{k} = 449\,\Omega \\ f_{H1} &= \frac{1}{2\pi(449)(383 \times 10^{-12})} = \mathbf{926}\,\text{kHz} \end{aligned}$$

### 6.5 Output-side cutoff

$$C_{out} = C_\mu\left(1 + \tfrac{1}{|A_m|}\right) + C_W \approx C_\mu + C_W, \qquad R_{th,out} = R_C \parallel R_L, \qquad \boxed{f_{H2} = \frac{1}{2\pi R_{th,out} C_{out}}} \tag{17}$$

$$C_{out} = 4.05 + 10 = 14.0\,\text{pF}, \qquad f_{H2} = \frac{1}{2\pi(2000)(14.0 \times 10^{-12})} = \mathbf{5.67}\,\text{MHz}$$

### 6.6 Which cutoff wins?

Going *up* in frequency from mid-band, the first thing you hit is the **lowest** of the high-frequency cutoffs:

$$f_H \approx \min\left(f_{H1},\ f_{H2}\right) = \min(926\,\text{kHz},\ 5.67\,\text{MHz}) = \mathbf{926}\,\text{kHz} \tag{18}$$

(exact composite value: 904 kHz). The input network dominates, and it dominates *because of Miller multiplication*. Without Miller, $C_{in}$ would be only 34 pF and $f_{H1}$ would be 10.4 MHz — more than eleven times wider.

**Gain–bandwidth trade-off**

$f_{H1} \approx \dfrac{1}{2\pi R_{th,in} C_\mu |A_m|}$ when Miller dominates. Since $f_{H1}$ is inversely proportional to $|A_m|$, the product $|A_m| \times f_H$ is roughly constant for a given transistor and bias. Ask for more gain and you get proportionally less bandwidth. This is why high-frequency designs use low-gain stages in cascade, or the cascode configuration, which kills the Miller effect by preventing the collector of the input transistor from swinging.

# 7 The Complete Curve, and Why It Has That Shape

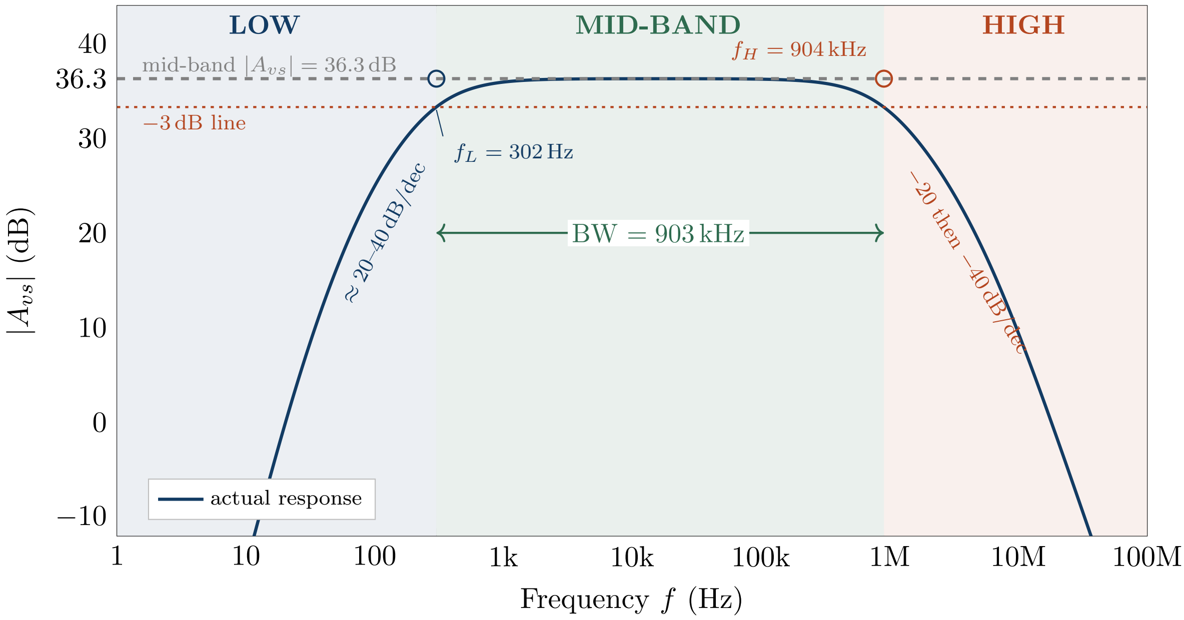



Figure 6: Complete gain–frequency response of the worked example. Computed exactly from the reactance formulas of Sections 5 and 6 — no approximations beyond treating each capacitor's network independently.

## 7.1 The curve is five simple curves added together

Figure 6 looks complicated, but it is built from five copies of the two elementary shapes you met in Figures 3 and 4. Because dB values *add* when gains multiply, the total response is simply

$$A_{dB}(f) = \underbrace{36.3\,\text{dB}}_{\text{mid-band level}} + \underbrace{d_1(f) + d_2(f) + d_E(f)}_{\text{drops from the three large capacitors}} + \underbrace{d_{in}(f) + d_{out}(f)}_{\text{drops from the small capacitors}}$$

where each $d$ is a negative number of decibels outside its own band and exactly zero in mid-band.

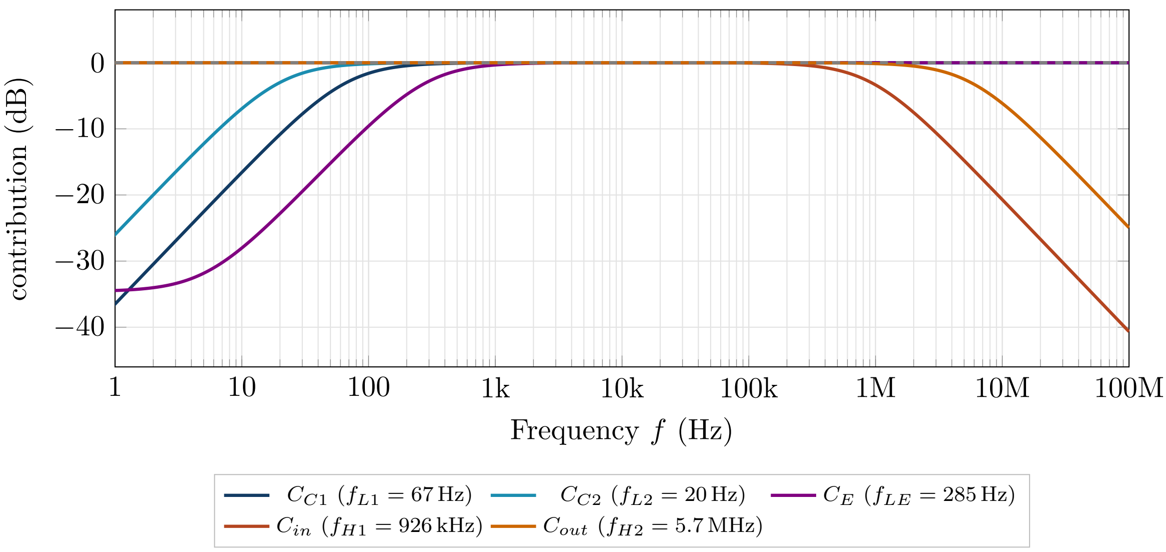



Figure 7: The five individual contributions. Each is flat at 0 dB (no effect) through mid-band and bends down by 20 dB/decade beyond its own corner frequency. Adding all five to the mid-band level of 36.3 dB reproduces Figure 6 exactly.

Three things are worth staring at in Figure 7:

- **Every curve is flat at 0 dB across the middle.** That is what "mid-band" means, and it is why the total is flat there.
- **The curve that bends first as you move outward is the one that sets the cutoff.** Moving left, $C_E$ (violet) departs from 0 dB well before the other two. Moving right, $C_{in}$ (red) departs well before $C_{out}$. Everything else is a secondary correction.
- **$C_E$'s curve flattens out again below 5 Hz.** It cannot pull the gain down forever, because once $X_{C_E}$ exceeds $R_E$ itself there is nothing left to un-bypass. The coupling capacitors, by contrast, fall without limit.

## 7.2 Reading the shape, region by region

**The flat middle.** Between roughly 300 Hz and 900 kHz the gain sits at a constant 36.3 dB. It is flat because in this window *no capacitor is doing anything.* The large ones have reactances far below the surrounding resistances (they are wires); the small ones have reactances far above (they are absent). The circuit is purely resistive, and resistors do not care about frequency. **A flat region on a gain plot is always a region where every capacitor has been resolved into either a short or an open.**

**The fall at the left.** Going down in frequency, $X_C = 1/2\pi fC$ of the large capacitors grows. The first one to matter is the one with the highest cutoff — here $C_E$ at 285 Hz. As $C_E$ loses its grip, $R_E$ becomes progressively un-bypassed, negative feedback returns, and the gain slides toward $-R'_C/(r_e + R_E)$. Continuing down, $C_{C1}$ (at 67 Hz) and $C_{C2}$ (at 20 Hz) start dropping signal voltage across themselves instead of delivering it. Each active capacitor contributes another 20 dB/decade of slope, so the curve steepens as you descend. Because the coupling capacitors are genuinely in series with the signal, the gain heads to $-\infty$ dB (zero) at DC — as it must, since the amplifier is DC-blocked by construction.

**The fall at the right.** Going up in frequency, $X_C$ of the tiny internal capacitors shrinks. Once it becomes comparable to the surrounding Thevenin resistance, those capacitors start short-circuiting the signal to ground before it can be amplified. The input network gives out first (926 kHz), largely because Miller multiplication has inflated 4 pF into 353 pF. The output network follows at 5.7 MHz, adding a second 20 dB/decade and steepening the curve to 40 dB/decade. Physically, the transistor's stored charge cannot be moved fast enough to follow the input.

**Why the two ends are mirror images.** Both ends are governed by the same right-triangle divider, Eq. (2). At the low end the output is taken across the resistor and the useless voltage appears across the capacitor; at the high end the roles swap. Same mathematics, opposite orientation — hence the symmetry of Figures 3 and 4.

**Why the corners are rounded, not sharp.** The straight dashed asymptotes cross at $f_c$, but the true curve is 3 dB below that corner. There is no sudden event at $f_c$: the capacitor's reactance changes smoothly, so the transition from "acting as a short" to "acting as an open" is gradual. The corner frequency is a label we attach to the middle of a smooth handover, not a switch being thrown.

**The shape in one sentence**

The curve is flat wherever every capacitor is unambiguously a short or an open, and it falls at 20 dB per decade for each capacitor that is currently in its ambiguous middle region — the big capacitors going ambiguous as you move left, the small ones as you move right.

### 7.3 A note on phase (optional)

At the low end, the series capacitors make the output *lead* the input by up to a further 90° per capacitor; at the high end, the shunt capacitors make it *lag* by up to 90° per capacitor. At each cutoff frequency the extra shift is exactly 45°. In mid-band the only phase shift is the 180° inversion of the CE stage. You do not need this for the gain curve, but it matters when such a stage is placed inside a feedback loop.

## 8 Worked Example: Complete Summary

| Quantity | Formula | Value | Comment |
|---|---|---|---|
| $I_E$ | $(V_B - 0.7)/R_E$ | 1.13 mA | bias |
| $r_e$ | $V_T/I_E$ | 23 Ω | |
| $g_m$ | $1/r_e$ | 43.6 mS | |
| $r_\pi$ | $\beta r_e$ | 2.29 kΩ | |
| $R'_C$ | $R_C \parallel R_L$ | 2.00 kΩ | |
| $R_i$ | $R_1 \parallel R_2 \parallel r_\pi$ | 1.78 kΩ | |
| $A_m$ | $-g_m R'_C$ | −87.2 (38.8 dB) | base to output |
| $A_{vs}$ | $A_m R_i/(R_i + R_s)$ | −65.2 (36.3 dB) | source to output |
| $f_{L1}$ | $1/2\pi(R_s + R_i)C_{C1}$ | 66.8 Hz | input coupling |
| $f_{L2}$ | $1/2\pi(R_C + R_L)C_{C2}$ | 19.9 Hz | output coupling |
| $f_{LE}$ | $1/2\pi[R_E \parallel (r_e + R'_s/(\beta+1))]C_E$ | **284.8** Hz | **dominant** |
| $f_L$ | $\approx \max(\cdot)$; exact | 285 Hz; 302 Hz | |
| $C_{in}$ | $C_\pi + C_\mu(1 + \lvert A_m \rvert)$ | 383 pF | 353 pF of it is Miller |
| $R_{th,in}$ | $R_s \parallel R_1 \parallel R_2 \parallel r_\pi$ | 449 Ω | |
| $f_{H1}$ | $1/2\pi R_{th,in} C_{in}$ | **926** kHz | **dominant** |
| $C_{out}$ | $C_\mu(1 + 1/\lvert A_m \rvert) + C_W$ | 14.0 pF | |
| $f_{H2}$ | $1/2\pi(R_C \parallel R_L)C_{out}$ | 5.67 MHz | |
| $f_H$ | $\approx \min(\cdot)$; exact | 926 kHz; 904 kHz | |
| BW | $f_H - f_L$ | 903 kHz | |
| GBW | $\lvert A_{vs} \rvert \cdot$BW | 58.9 MHz | roughly constant |

### 8.1 Design lessons that fall straight out of the table

1. **To lower $f_L$:** increase $C_E$ first — it is the bottleneck, and it sees the smallest resistance. Increasing $C_{C1}$ or $C_{C2}$ instead would be wasted effort.
2. **To raise $f_H$:** reduce $R_{th,in}$ (drive the stage from a lower-impedance source), reduce the stage gain, or eliminate the Miller effect entirely with a cascode.
3. **Gain and bandwidth trade against each other** through the Miller term. You cannot have both from a single CE stage.
4. **The dominant capacitor at each end is the one seeing the extreme resistance:** the smallest $R_{eq}$ at the low end (highest $f_c$), and the largest $R \cdot C$ product at the high end (lowest $f_c$).

## 9 Common Misconceptions

**Six mistakes that cost marks**

1. **"The lower cutoff is the sum of the three."** No. Cutoffs are not added. The highest one dominates; the true $f_L$ is slightly *above* the highest, never the sum.
2. **"Bigger capacitors always improve the response."** Only at the low end, and only for the dominant one. The internal $C_\pi$ and $C_\mu$ are capacitors too, and making them bigger would be a disaster.
3. **"$C_E$ makes the gain fall to zero at DC."** It does not; it makes the gain fall to the un-bypassed value $-R'_C/(r_e + R_E)$. The coupling capacitors are what force zero gain at DC.
4. **"$C_\mu$ is only 4 pF, so it is negligible."** It is the single most important capacitor in the circuit, because Miller multiplication scales it by $(1 + |A_v|)$.
5. **"Use $R_E$ as the resistance seen by $C_E$."** You must use $R_E$ in *parallel* with the resistance looking into the emitter, which is far smaller than $R_E$ and dominates the parallel combination.
6. **"The corner is where the gain suddenly drops."** The corner is a smooth 3 dB handover. The asymptotes are a drawing convenience, not physics.

**Self-test**

1. In the example, $C_E$ is changed from 20 $\mu$F to 100 $\mu$F. What is the new $f_{LE}$, and what is the new overall $f_L$ (approximately)?
2. The load $R_L$ is removed (open circuit). What happens to $A_m$, to $f_{L2}$, and to $f_{H2}$?
3. A student replaces the transistor with one having $C_\mu = 8$ pF instead of 4 pF, everything else unchanged. Estimate the new $f_{H1}$.
4. Explain, without any formula, why a stage with a gain of 500 has a narrower bandwidth than an otherwise identical stage with a gain of 20.
5. Why can the low-frequency analysis safely ignore $C_\pi$ and $C_\mu$?

**Answers**

1. $f_{LE} = 284.8 \times (20/100) = 57.0$ Hz. Now $C_{C1}$ at 66.8 Hz becomes the highest, so $f_L \approx 67$ Hz (exact composite a little above, around 85 Hz). Note the diminishing return: you cannot go below 67 Hz without also enlarging $C_{C1}$.
2. $R'_C = R_C = 4\,\text{k}\Omega$, so $|A_m|$ *doubles* to 174. $f_{L2}$ becomes irrelevant (no path through $C_{C2}$). $f_{H2}$ halves to about 2.8 MHz because $R_{th,out}$ doubled — and $f_{H1}$ also falls, roughly halving, because the Miller capacitance doubled with the gain. Removing the load buys gain and pays for it in bandwidth.
3. $C_{in} = 30 + 8(88.2) = 736$ pF, so $f_{H1} = 1/(2\pi \cdot 449 \cdot 736\text{p}) \approx 482$ kHz — roughly halved.
4. Higher gain means the collector swings further for a given base movement, so the voltage across the base–collector capacitance swings further, so the source must supply more charge, so the input looks more capacitive, so the input network gives out at a lower frequency.
5. Because at low frequencies their reactance is in the megohms or higher — thousands of

times larger than any resistance in the circuit — so no appreciable current flows through them.

## A Appendix A — Where the $\sqrt{R^2 + X_C^2}$ Comes From

You may take this on trust, but here is the reason, using nothing beyond sinusoids.

Apply $v(t) = V_m \sin(2\pi f t)$ to a series $RC$ pair carrying current $i(t) = I_m \sin(2\pi f t)$. The voltage across the resistor is $v_R = I_m R \sin(2\pi f t)$ — in step with the current. The voltage across the capacitor is set by the charge it has accumulated, which is the running total of the current; the running total of a sine is a negative cosine, so $v_C = I_m X_C \sin(2\pi f t - 90°)$ — a quarter cycle behind.

So the two voltages never peak at the same instant: when one is at maximum the other is at zero. Their sum is therefore *not* $I_m R + I_m X_C$. Adding two sinusoids of the same frequency that are 90° apart gives a sinusoid whose amplitude is the hypotenuse of a right triangle with the two amplitudes as legs:

$$V_m = \sqrt{(I_m R)^2 + (I_m X_C)^2} = I_m \sqrt{R^2 + X_C^2}.$$

Everything in this document follows from that one geometric fact. (If you later study phasors, you will recognise this as $|Z| = |R + 1/j\omega C|$ — but you do not need that notation to use it.)

## B Appendix B — Formula Sheet

| | |
|---|---|
| Reactance | $X_C = 1/(2\pi f C)$ |
| Cutoff of any single $RC$ section | $f_c = 1/(2\pi R_{eq} C)$ |
| High-pass magnitude (low end) | $1/\sqrt{1 + (f_c/f)^2}$ |
| Low-pass magnitude (high end) | $1/\sqrt{1 + (f/f_c)^2}$ |
| Decibels | $A_{dB} = 20 \log_{10} \lvert A_v \rvert$ |
| Mid-band gain | $A_m = -g_m (R_C \parallel R_L) = -(R_C \parallel R_L)/r_e$ |
| Input resistance | $R_i = R_1 \parallel R_2 \parallel r_\pi$ |
| Input coupling | $f_{L1} = 1/[2\pi(R_s + R_i) C_{C1}]$ |
| Output coupling | $f_{L2} = 1/[2\pi(R_C + R_L) C_{C2}]$ |
| Emitter bypass | $f_{LE} = 1/[2\pi(R_E \parallel (r_e + R_s'/(\beta{+}1))) C_E]$ |
| Overall | $f_L \approx \max(f_{L1}, f_{L2}, f_{LE})$ |
| Miller input capacitance | $C_M = C_\mu (1 + \lvert A_v \rvert)$ |
| Input network | $f_{H1} = 1/[2\pi(R_s \parallel R_1 \parallel R_2 \parallel r_\pi)(C_\pi + C_M)]$ |
| Output network | $f_{H2} = 1/[2\pi(R_C \parallel R_L)(C_\mu + C_W)]$ |
| Overall | $f_H \approx \min(f_{H1}, f_{H2})$ |
| Bandwidth | $\mathrm{BW} = f_H - f_L$ |
| Slope per active capacitor | 20 dB per decade |

*End of document.*